Free DSA Roadmap (0 to Advanced)

Phase 1: Basics

Arrays: Linear Search, Largest, Second Largest, Sorted, Remove Duplicates, Rotate, Move Zeros, Union, Missing Number, Consecutive Ones, Single Number, Longest Subarray Sum K, Two Sum, Kadane.

Phase 2: Sorting

Bubble, Selection, Insertion, Merge, Quick, Count Inversions, Merge Intervals, Sort Colors.

Phase 3: Binary Search

Binary Search, Lower/Upper Bound, Search Insert, Floor/Ceil, First/Last Occurrence, Rotated Array, Peak, Square Root, Nth Root, Koko, Ship Packages, Split Array.

Phase 4: Strings

Reverse, Palindrome, Anagram, LCP, Roman, Compression, Reverse Words, atoi, Longest Palindrome, Longest Unique Substring, Minimum Window, Group Anagrams, KMP, Rabin-Karp, etc.

Phase 5: Linked List

Reverse, Middle, Detect Cycle, Merge, Remove Nth, Add Two Numbers, Reverse K Group, Clone, LRU Cache.

Phase 6: Stack & Queue

Valid Parentheses, Min Stack, NGE, Stock Span, Histogram, Sliding Window Max, Queue/Stack implementations.

Phase 7: Recursion

Factorial, Fibonacci, Power, Subsets, Subsequences, Combination Sum, Permutations, N-Queens, Sudoku.

Phase 8: Trees

Traversals, Height, Diameter, Balanced, Views, LCA, Serialize, Morris, Flatten.

Phase 9: BST

Search, Insert, Delete, Kth Smallest, Validate, Recover, Iterator.

Phase 10: Heap

Heap, Heap Sort, K Largest, Top K Frequent, Median Finder.

Phase 11: Graphs

BFS, DFS, Islands, Flood Fill, Cycle, Topo Sort, Bipartite, Dijkstra, Bellman-Ford, MST, DSU.

Phase 12: DP

Fibonacci, Climbing Stairs, Knapsack, Coin Change, LCS, LIS, Edit Distance, MCM, Burst Balloons.

Phase 13: Greedy

Activity Selection, Job Sequencing, Fractional Knapsack, Huffman, Jump Game.

Phase 14: Tries

Trie, Prefix, Word Dictionary, Maximum XOR.

Phase 15: Bit

XOR, Power of Two, Counting Bits, Single/Missing Number.

Daily Plan

30 min revision, 45 min learning, 60–90 min solving 3–5 problems, 15 min review.